CLASS 4 · PHASE 1 · ANALYSIS · BRICK 4 CLOSES PHASE 1

Statistics Refresher: Letting the Time Log Speak



CS 251 — Integrative Project 1 · Class 4 · Friday, October 16, 2026

1-hour lecture in four parts · 2-hour workshop (Brick 4) · 2 hours at home

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C4

Where we are

- Class 1 — Boréal Studio, the toolbox, your 14-day time log
- Class 2 — problem statement, research questions, requirements
- Class 3 — a price for every routine: about **5 821 $** a month; e-mail triage about **1 640 $** (28.2 %)
- Every one of those dollars is **minutes × hourly value**
- Today: what is typical · how much runs vary · which runs are unusual · what the log can prove
- **Brick 4** — the statistical analysis script — closes **Phase 1**

1 640 $

per month: manual e-mail triage (Class 3)

28.2 %

of the studio's routine bill

C4

What you can do by the end of this class

- **Compute** the mean, median, mode, range, sample variance, **SD** (Standard Deviation) and **CV** (Coefficient of Variation) — by hand and with Python's `statistics` module
- **Read and compare**: build a histogram, flag outliers with a rule stated in advance, read a scatter plot and r, compare two groups honestly
- **State the limits** of a 14-day, five-person data set next to every conclusion

RQ1 (Research Question 1) of the project

Which routines cost the most time, and which are the most variable?

Today answers it with statistics.

C4

The hour at a glance

0:00–0:11
Part 1
Lesson 4.1 — Central tendency

0:11–0:15
Activity 1
dot plot + fun quiz

0:15–0:26
Part 2
Lesson 4.2 — Dispersion

0:26–0:30
Activity 2
s by hand + REPL check

0:30–0:41
Part 3
Lesson 4.3 — Distributions and outliers

0:41–0:45
Activity 3
zen break + flash cards

0:45–0:56
Part 4
Lesson 4.4 — Relationships, groups and limits

0:56–1:00
Activity 4
fun quiz

- Then: **2-hour workshop** — Brick 4 “Statistical Analysis Script” · then **2 hours at home**

C4

The data of the day

routine	category	runs in the file
email_triage	communication	55 manual + 15 jarvis
standup_notes	communication	70 manual
scheduling	planning	70 manual
invoice_check	finance	70 manual
social_post	marketing	70 manual
daily_report	reporting	35 manual + 35 template

- 14 working days (September 14 – October 1, 2026) × 5 staff × 6 routines = **420 rows**
- **Rule of the day: describe one process at a time** — filter by routine and tool first

4.1

4.1 Why “typical” needs a definition

- The owner asks: *how long does e-mail triage take?* — no single run answers it
- A **measure of central tendency** is one number that represents the centre of a set of values
- Three standard measures: **mean**, **median**, **mode**
- When they disagree, some values lie far from the others — that is information
- Notation: **n** = number of runs · $\mathbf{x_1}$ … $\mathbf{x_n}$ = minutes · **Σ** = “add up”

Lesson 4.1 target

Compute mean, median, mode and range, and choose which value to report as typical — with a reason.

4.1

The mean

$$\bar{x} = (x_1 + x_2 + \ldots + x_n) / n = \Sigma x / n$$

- **$\bar{x}$** (“x-bar”) is the mean of a sample
- It uses **every value**: one very large value changes it
- Python: `sum(values) / len(values)` or `statistics.mean(values)`
- $\Sigma x = n \cdot \bar{x}$ exactly — the mean is the measure to multiply into a monthly total

4.1

One week with a Monday backlog

Monday	Tuesday	Wednesday	Thursday	Friday
105	36	33	38	34

246
Σx (minutes)

49.2
$\bar{x}$ = 246 / 5

4 of 5
runs are below the mean

- S2's e-mail triage, one week (illustrative values)
- The mean is longer than every run except Monday's

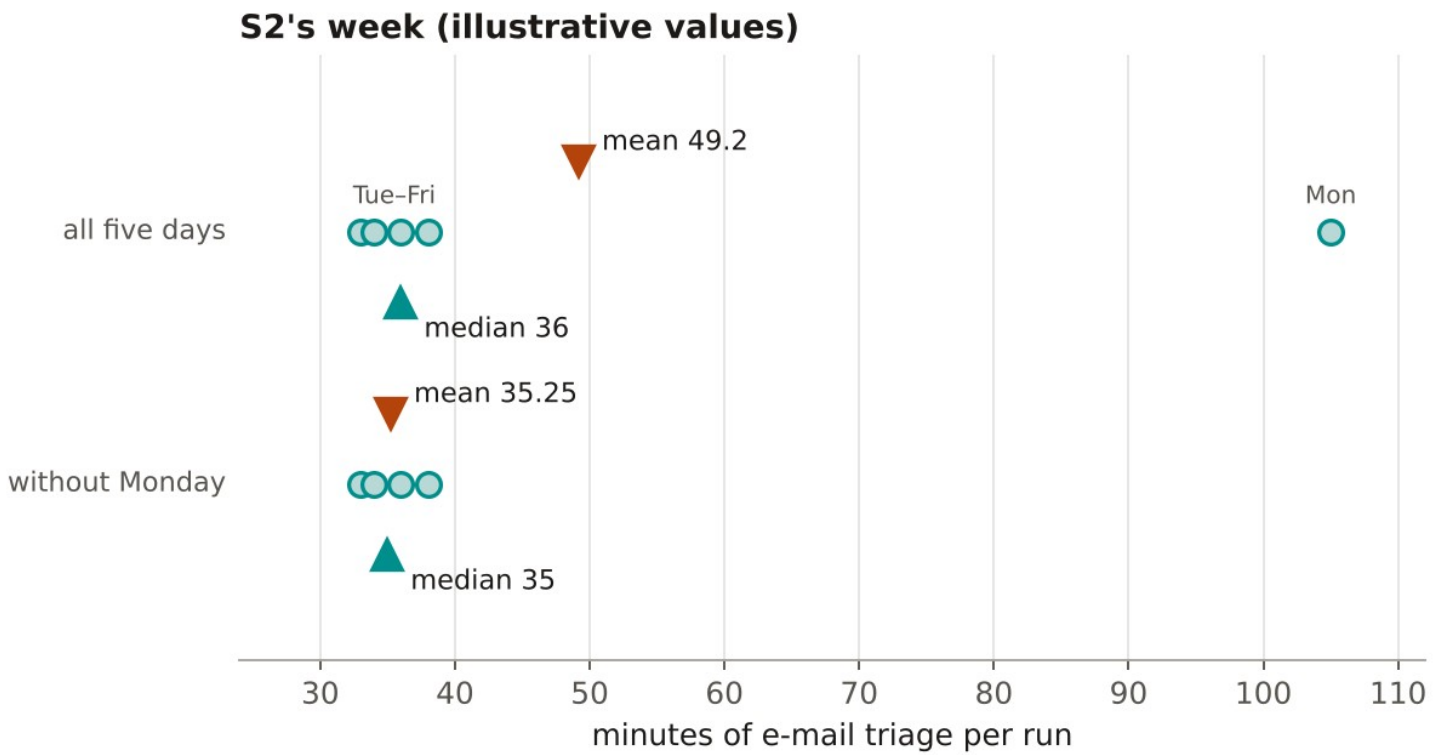
4.1

The median

- The **median** is the middle value of the **sorted** data
- Odd n: the value in position (n + 1) / 2
- Even n: the average of positions n/2 and n/2 + 1
- Week sorted: 33, 34, **36**, 38, 105 → median **36 minutes**
- Without Monday: 33, **34, 36**, 38 → (34 + 36) / 2 = **35 minutes**
- Code: `sorted(values)` (a copy), `n // 2`, `n % 2` — Python counts from 0

4.1

The median resists one extreme run



S2's week (illustrative): mean and median with and without Monday

- A measure that one extreme value cannot move far is **resistant** (robust)
- An **outlier** lies unusually far from the other values — Lesson 4.3 defines "unusually far"

4.1

The mode, and what to do with ties

- The **mode** is the most frequent value
- Natural for **discrete counts**: interruptions 5, 1, 0, 2, 1 → mode **1**
- Weak for measured times: 29, 32 and 37 minutes **tie** among the 55 manual e-mail runs
- Two modes: **bimodal**; more: **multimodal**
- `statistics.mode` → first mode met · `statistics.multimode` → all of them
- Write your tie rule in the docstring — StatsEngine returns the list

4.1

The range

72

minutes: 105 − 33

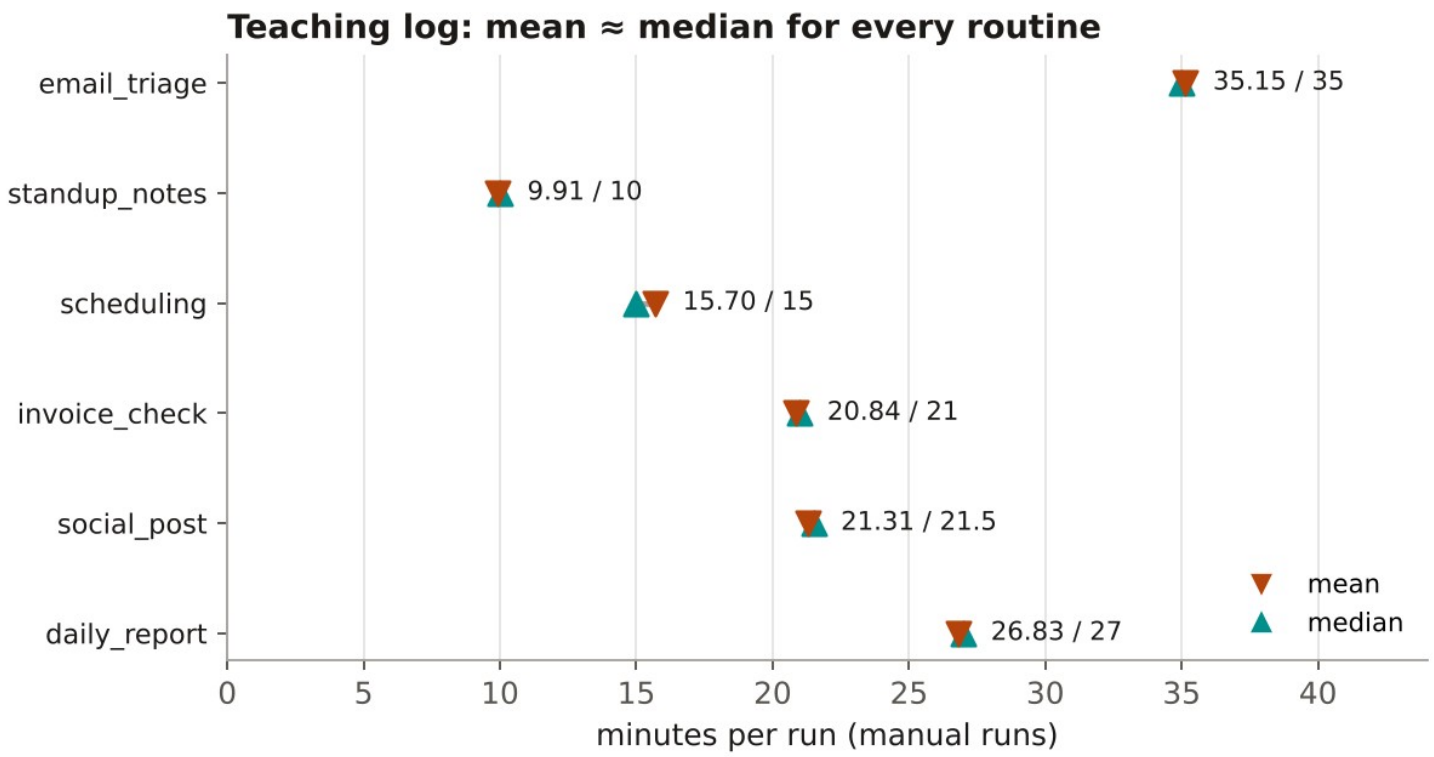
2

values decide the range

- **Range** = largest value − smallest value
- Quick to compute, easy to misread: one unusual Monday decides it
- It describes **spread**, not centre — Lesson 4.2 does better

4.1

Which number describes a typical run?



Mean and median per routine, manual runs of the teaching copy

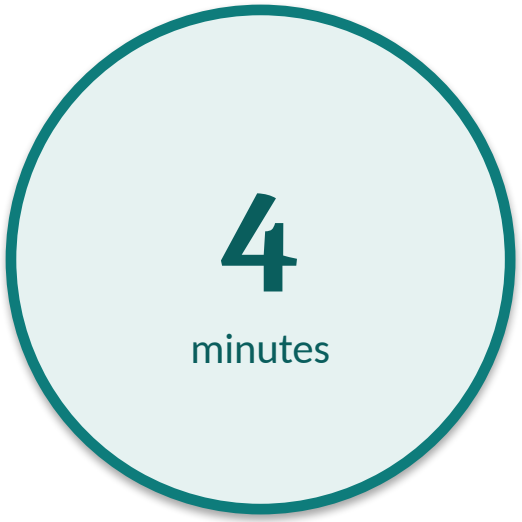
- **Median** — a few extreme values or a long tail
- **Mean** — symmetric data, and every number that will be multiplied
- **Both**, with the reason, when they differ by more than a few percent

4.1

Activity 1 — dot plot and fun quiz

⏱️ **0:11–0:15 · Visualization + fun quiz · 4 minutes**

- Draw S2's week as a dot plot (axis 30–110 minutes)
- Mark the **mean (49.2)** and the **median (36)**
- 1 — Which is the honest “typical day”: mean or median? Why?
- 2 — Monday's interruptions rise from 5 to 9. What happens to the mode?



4.1

Real-life example 4.1 — the owner's planning slot

TASK

Mean, median, mode and range of e-mail triage — for S2's week, then for all 55 manual runs — and the value for the owner's planning sheet, with a one-sentence reason.

Deliverables

Table routine · n · mean · median · mode(s) · range
One routine computed by hand
One sentence: the typical value and why

Where it goes next

Brick 4 (Oct 22) — descriptives per routine
Brick 6 (Nov 5) — StatsEngine.mean / median / mode
Brick 9 (Dec 3) — your first pytest value
Mid-Term Part A (Nov 20)

Demo in the lecture module: Start / Pause, Step, Restart — the trace window shows every executed line.

4.1

Videos for home — Lesson 4.1

11:23

Mean, Median, and Mode: Measures of Central Tendency: Crash Course Statistics #3

CrashCourse · YouTube · The three measures, why one extreme value moves the mean but not the median, and what skew does to them.

4:15

Impact on median and mean when increasing highest value | 6th grade | Khan Academy

Khan Academy · YouTube · One value is made larger: the mean rises, the median stays — the Monday-backlog effect in four minutes.

8:54

Statistics intro: Mean, median, and mode | Data and statistics | 6th grade | Khan Academy

Khan Academy · YouTube · A slow, worked introduction to the three measures for anyone who wants a full refresher.

All three were confirmed working (oEmbed) on October 11, 2026, and are 15 minutes or shorter. Check a link with VirusTotal before you open it.

4.2

4.2 Same centre, different reliability

Set A — mean 10

8, 10, 9, 12, 11
no run more than 2 minutes
from the mean
a planner can rely on 10

Set B — mean 10

4, 16, 6, 15, 9
runs from 4 to 16 minutes

10 is a poor guide

- **Dispersion** (spread, variability): how far values typically lie from their centre

4.2

Deviations always add up to zero

x	8	10	9	12	11	Σ
$d = x - \bar{x}$	−2	0	−1	+2	+1	0
d^2	4	0	1	4	1	10

- **Deviation** $d_i = x_i - \bar{x}$ (here $\bar{x} = 10$)
- $\Sigma d = 0$ for every data set — the mean is their balance point
- Squaring removes the sign: $\Sigma d^2 =$ **10**, the **sum of squares**

4.2

Sample variance and standard deviation

$$s^2 = \Sigma(x_i - \bar{x})^2 \,/\, (n - 1) \qquad s = \sqrt{(s^2)}$$

- Set A: $s^2 = 10 / 4 =$ **2.5 min²** → $s = \sqrt{2.5} \approx$ **1.58 minutes**
- Read s as: a run typically lies about 1.6 minutes from the mean
- Set B: $\Sigma d^2 = 114$, $s^2 = 28.5$, $s \approx$ **5.34 minutes**
- The variance is in **min²**; the **SD** (Standard Deviation) is back in **minutes**

4.2

Why n − 1? Sample versus population

Sample — divide by n − 1

statistics.variance / stdev
set A: s = 1.581
spreadsheet: STDEV.S
JARVIS's StatsEngine

Population — divide by n

statistics.pvariance / pstdev
set A: σ = 1.414
spreadsheet: STDEV.P
all values of interest observed

- n − 1 = the **degrees of freedom**: once $\bar{x}$ is fixed, only n − 1 deviations are free

4.2

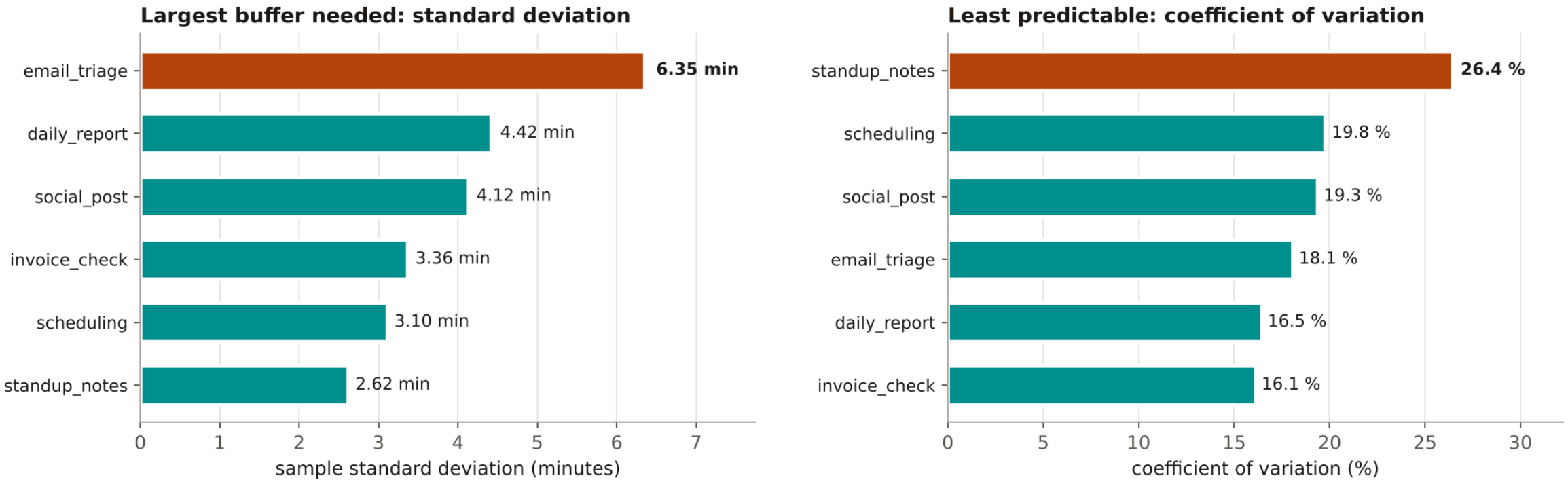
The coefficient of variation

$$CV = s / \bar{x} \quad (\times 100, \text{ in } \%)$$

- A standard deviation grows with the size of the values
- The **CV** (Coefficient of Variation) has no unit: it compares routines of different length
- Set A: 1.58 / 10 = **15.8 %** · set B: 5.34 / 10 = **53.4 %**
- Meaningful for data with a true zero (minutes, dollars, counts)

4.2

Our studio: SD and CV rank routines differently



Manual runs: sample SD in minutes (left) and CV in % (right)

- **Plan the day** → **SD** (Standard Deviation) in minutes: e-mail triage needs the largest buffer (6.35)
- **Compare processes** → **CV** (Coefficient of Variation): stand-up notes are least predictable (26.4 %)

4.2

Describe one process at a time

6.35
standard deviation of manual e-mail runs

2.46
standard deviation of jarvis e-mail runs

10.54
standard deviation of both mixed

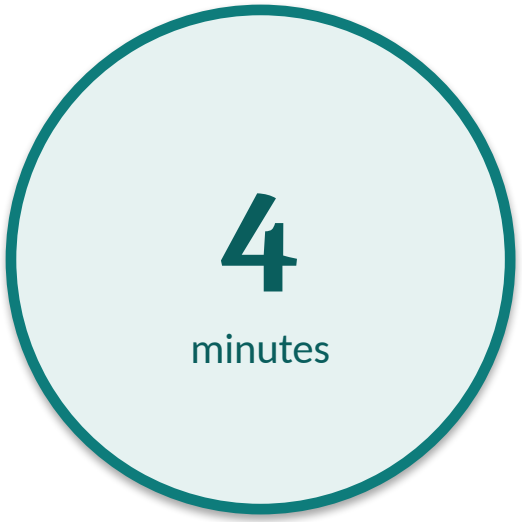
- The mixed **SD** (Standard Deviation) is larger than either group's **SD**
- The difference between the group means is counted as variation
- Filter by routine **and** tool first

4.2

Activity 2 — s by hand, then the REPL

⏱️ **0:26–0:30 · Short practical · 4 minutes**

- Five scheduling runs of S5: **14, 17, 12, 16, 16** minutes
- By hand: mean, deviations, sum of squares, s^2, s, **CV** (Coefficient of Variation)
- Check: `statistics.stdev([14, 17, 12, 16, 16])` and `statistics.pstdev(…)`
- Which one matches your hand result — and why is the other smaller?



4.2

Real-life example 4.2 — time buffers

TASK

SD (Standard Deviation) and **CV** (Coefficient of Variation) per routine (manual runs), two rankings, and a planning slot ⌈mean + **SD**⌉ per routine checked against the log.

Deliverables

Table n · mean · **SD** (Standard Deviation) · **CV** (Coefficient of Variation) · slot · % covered
Two sentences: largest **SD** (Standard Deviation), largest **CV** (Coefficient of Variation), and what each means
Set A by hand (for the tests)

Where it goes next

Brick 4 (Oct 22) — variance, **SD** (Standard Deviation), **CV** (Coefficient of Variation) per routine
Brick 6 (Nov 5) — StatsEngine with n − 1
Brick 9 (Dec 3) — oracle test against statistics.stdev
Brick 11 (Dec 17) — results tables

Demo in the lecture module: Start / Pause, Step, Restart — the trace window shows every executed line.

4.2

Videos for home — Lesson 4.2

11:47

Measures of Spread: Crash Course Statistics #4

CrashCourse · YouTube · Range, **IQR** (Interquartile Range), variance from squared deviations and the standard deviation, including the n − 1 of the sample variance.

7:04

What is the Coefficient Of Variation?? (+ examples!)

zedstatistics · YouTube · **CV** (Coefficient of Variation) = **SD** (Standard Deviation) ÷ mean, and why it is the right tool to compare spreads measured on different scales.

14:22

Calculating the Mean, Variance and Standard Deviation, Clearly Explained!!!

StatQuest with Josh Starmer · YouTube · The clearest explanation of population (÷ n) versus sample (÷ n − 1) — watch it at home before Brick 6.

All three were confirmed working (oEmbed) on October 11, 2026, and are 15 minutes or shorter. Check a link with VirusTotal before you open it.

4.3

4.3 From a list to a frequency table

bin (minutes)	25–29	30–34	35–39	40–44	45–49	50–54	total
runs	13	14	14	10	3	1	55
relative	23.6 %	25.5 %	25.5 %	18.2 %	5.5 %	1.8 %	100 %

- A **distribution**: which values occur, and how often
- **Bins** of equal width; no overlap; every value in exactly one bin (25 ≤ x < 30)
- **Relative frequency** = count / n

4.3

Histograms

- A **histogram** draws the frequency table: one bar per bin, bars touching
- Not a **bar chart**: its axis is a continuous numeric scale
- **Bin width** too wide hides the shape; too narrow shows noise
- About 50 values: start with 6–10 bins, then try wider and narrower
- Brick 4: a text histogram, one # per run · Brick 11: `matplotlib.pyplot.hist`

Text histogram (Brick 4)

```
25-29   ############ 13
30-34   ############## 14
35-39   ############## 14
40-44   ########## 10
45-49   ### 3
50-54   # 1
```

4.3

Shape: symmetric, skewed, bimodal

- **Symmetric**: halves are mirror images; mean ≈ median
- **Right-skewed**: long right tail; **mean > median** — typical of durations
- **Left-skewed**: the long tail is on the side of small values
- Manual e-mail runs: weak right skew (mean 35.15, median 35)
- **Bimodal**: add the 15 jarvis runs (10–19 minutes) → two peaks = two processes in one list

4.3

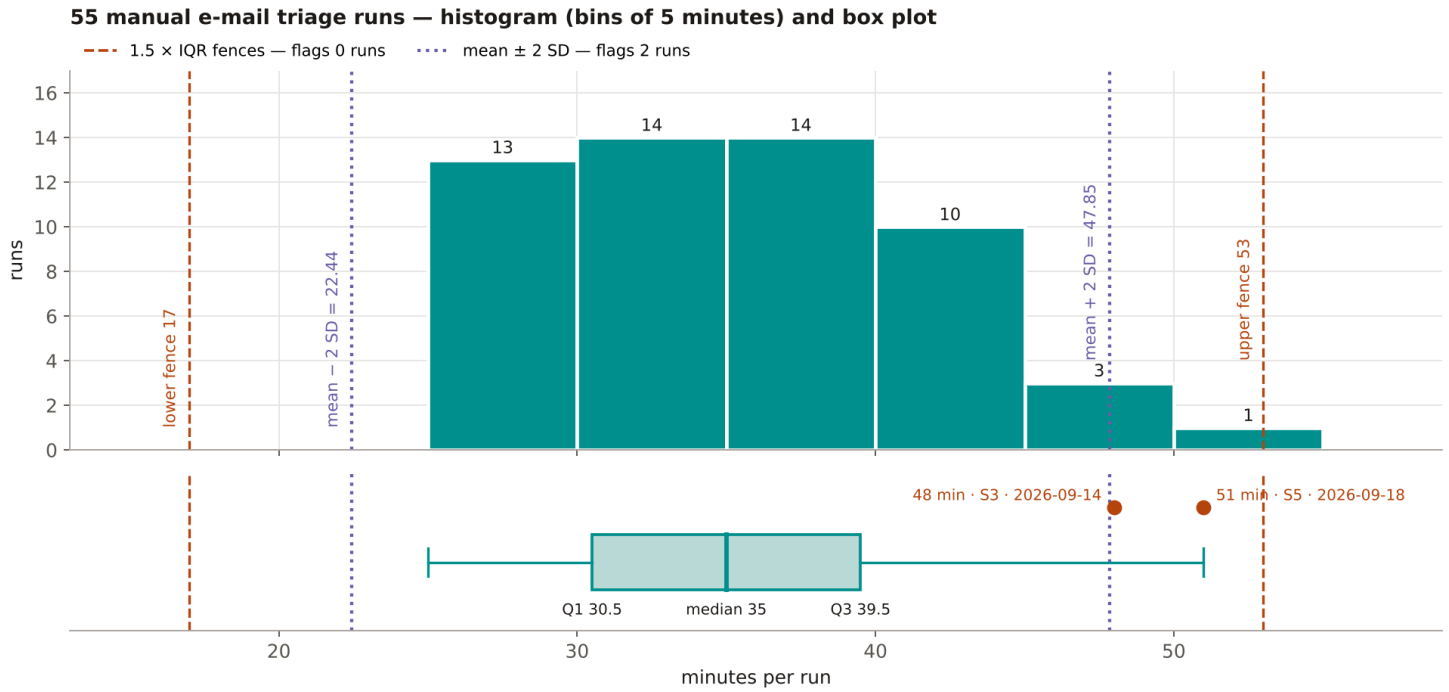
Quartiles and the IQR

$$\text{position} = p \times (n - 1) \quad \cdot \quad \text{IQR} = Q3 - Q1$$

- **Q1** (first quartile) ≈ 25 % below · **Q2** (second quartile) = median · **Q3** (third quartile) ≈ 75 % below
- **Q1** (first quartile): 0.25 × 54 = 13.5 → between 30 and 31 → **Q1 = 30.5**
- **Q3** (third quartile): 0.75 × 54 = 40.5 → between 39 and 40 → **Q3 = 39.5**
- **IQR (Interquartile Range) = 9 minutes**: the width of the middle half; resistant like the median
- `statistics.quantiles(values, n=4, method="inclusive")` — the default method differs: state yours

4.3

The box plot



Histogram and box plot of the 55 manual e-mail triage runs

- **Box plot**: box **Q1** (first quartile)–**Q3** (third quartile), median line, **whiskers** to the most extreme non-outlying values

4.3

Two outlier rules, two answers

1.5 × IQR (Interquartile Range) rule (fences)

Q1 (first quartile) − 1.5·**IQR** (Interquartile Range) = 17 minutes
Q3 (third quartile) + 1.5·**IQR** (Interquartile Range) = 53 minutes
flags 0 runs
built on resistant quartiles

mean ± 2 SD (Standard Deviation) rule

35.15 ± 2 × 6.35
= [22.44, 47.85]
flags 2 runs: 48 and 51 minutes
built on mean and **SD** (Standard Deviation)

- Choose and write the rule **before** you look at what it flags

4.3

What to do with a flagged run

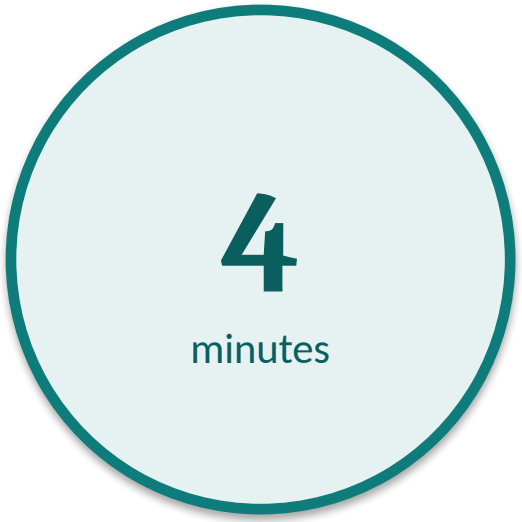
- Check it in the log first: 51 minutes = 3 interruptions · 48 minutes = Monday, first day
- 1 — **Keep** it: a real run (the default)
- 2 — **Correct** it: only a proven entry error, recorded in the bug log
- 3 — **Report separately**: results with and without it
- **Never delete silently** — e-mail triage would look cheaper than it is

4.3

Activity 3 — zen break and flash cards

⏱️ **0:41–0:45 · Zen break + flash cards · 4 minutes**

- **Zen break (2 minutes)**: stand, roll the shoulders, breathe 4-4-4 ten times
- **Flash cards in pairs (2 minutes)**: one reads the term, the other defines it
- Cards: mean · median · mode · s (n − 1) · **IQR** (Interquartile Range) · outlier rule



4.3

Real-life example 4.3 — checking the log

TASK

Frequency table and text histogram of the 55 manual e-mail runs, **Q1** (first quartile), **Q3** (third quartile), **IQR** (Interquartile Range) and fences, both outlier rules, and a decision for every flagged run.

Deliverables

Frequency table and text histogram
Q1 (first quartile), **Q3** (third quartile), **IQR** (Interquartile Range) and fences, method named
Flagged runs, a decision each, the one-sentence rule

Where it goes next

Brick 4 (Oct 22) — histogram and outlier rule
Brick 6 (Nov 5) — StatsEngine.outliers
Brick 9 (Dec 3) — edge-case tests (empty, one value, **IQR** (Interquartile Range) = 0)
Brick 11 (Dec 17) — histogram chart in the Reporter

Demo in the lecture module: Start / Pause, Step, Restart — the trace window shows every executed line.

4.3

Videos for home — Lesson 4.3

11:23

The Shape of Data: Distributions: Crash Course Statistics #7

CrashCourse · YouTube · Symmetric, right- and left-skewed, unimodal and bimodal distributions, and how a long tail moves the mean.

8:20

Judging outliers in a dataset | Summarizing quantitative data | AP Statistics | Khan Academy

Khan Academy · YouTube · **Q1** (first quartile), **Q3** (third quartile) and the **IQR** (Interquartile Range) computed by hand, the 1.5 × **IQR** fences applied, and the box plot drawn.

3:42

StatQuest: Histograms, Clearly Explained

StatQuest with Josh Starmer · YouTube · Bins, stacked counts and the effect of the bin width, in under four minutes.

All three were confirmed working (oEmbed) on October 11, 2026, and are 15 minutes or shorter. Check a link with VirusTotal before you open it.

4.4

4.4 Scatter plots

- A **scatter plot**: one point per run, two numeric variables
- x = **explanatory variable** (interruptions) · y = **response variable** (minutes)
- Read the **direction** (rising, falling), the **form** (line, curve, none) and the **strength**
- **Jitter**: a small horizontal shift that keeps points with the same x visible

Read three properties

Direction — rising or falling
Form — line, curve or none
Strength — how closely the points follow the form

4.4

Pearson's correlation coefficient r

$$r = \Sigma(x_i - \bar{x})(y_i - \bar{y}) \;/\; \sqrt{[\,\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2\,]}$$

- **Pearson's r**: direction and strength of the **linear** relationship
- +1 rising line · −1 falling line · ≈ 0 no linear relationship
- No unit; swapping x and y does not change r
- Python: `statistics.correlation(xs, ys)` (Python 3.10 and later)

4.4

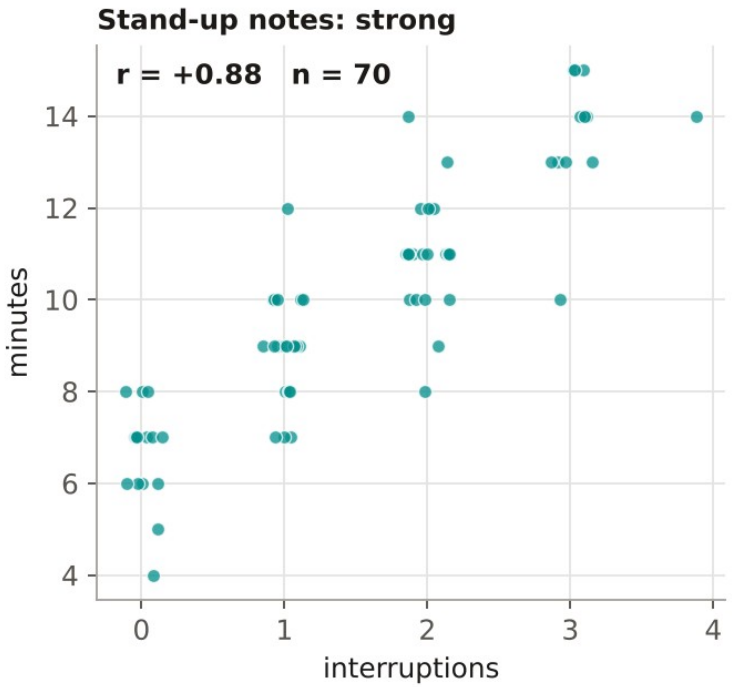
Reading the size of r

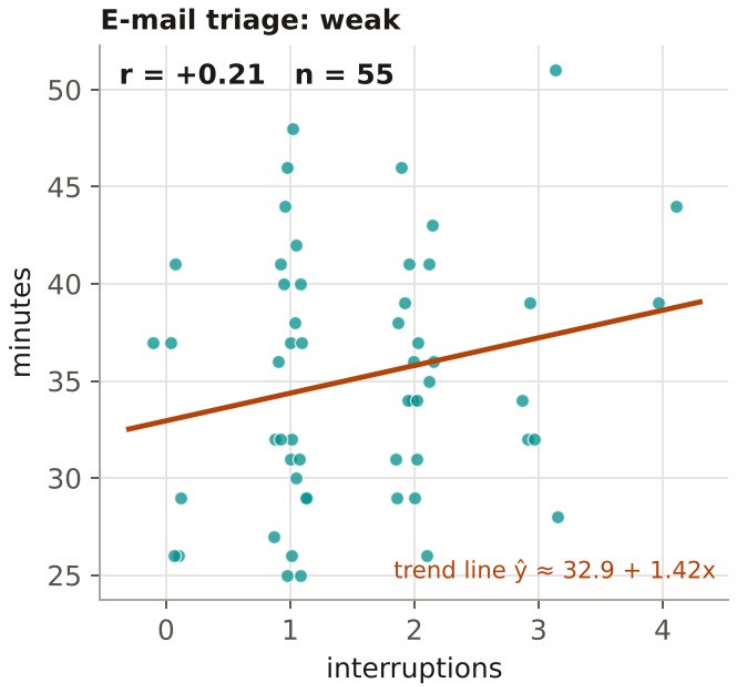
absolute value of r	below 0.10	0.10–0.29	0.30–0.69	0.70 and above
wording	negligible	weak	moderate	strong

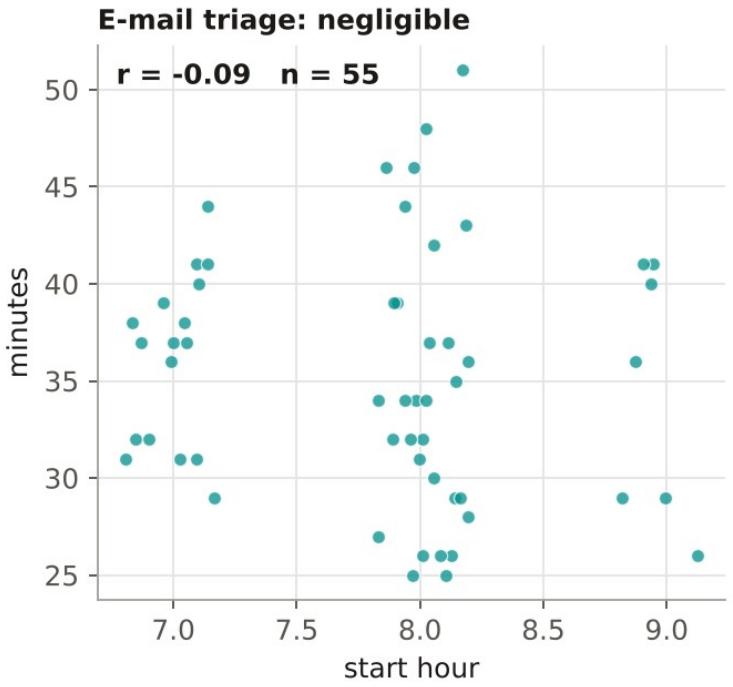
- Always give the **sign**, the **value**, **n** and the **period**
- “r = +0.21 between interruptions and minutes, 55 manual e-mail runs, September 14 – October 1, 2026”

4.4

Our studio's three scatter plots







Manual runs, jittered x: r = +0.883, +0.215 and −0.093

- **r measures how closely the points follow a line**, not the size of the effect

4.4

The trend line answers a different question

1.42
minutes per interruption (e-mail)

2.21
minutes per interruption (stand-up)

0.21 / 0.88
r for e-mail / stand-up

- E-mail trend line: **$\hat{y} \approx 32.9 + 1.42\,x$**
- Least squares and `statistics.linear_regression` — Class 7

4.4

Before and after the daily-report template

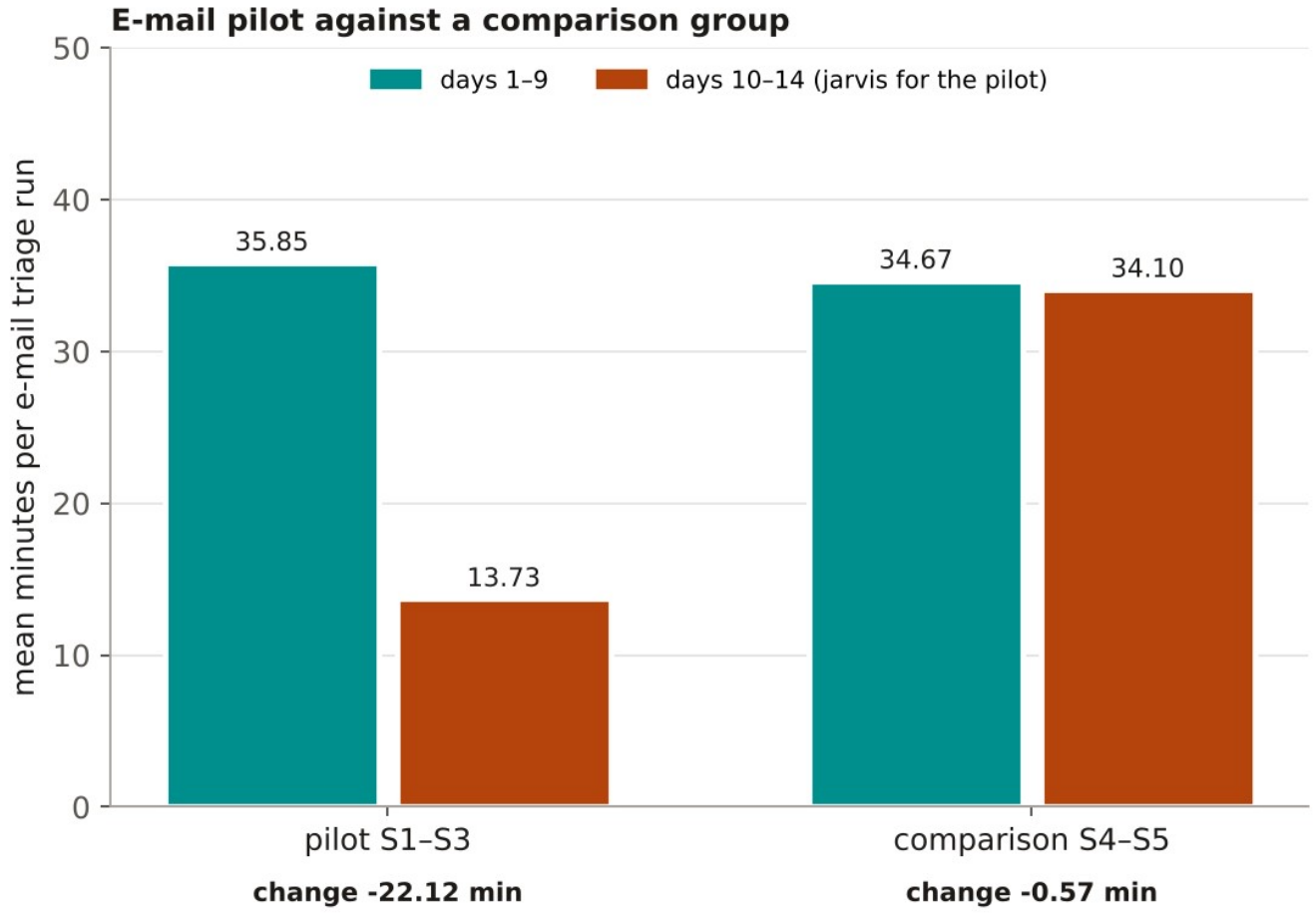


Daily report: mean minutes per run, ± 1 SD

- Days 1–7 manual: **26.83** (**SD** (Standard Deviation) 4.42, n 35) · days 8–14 template: **16.83** (**SD** 3.06, n 35)
- Always give **n and the SD (Standard Deviation)** next to a mean

4.4

The e-mail pilot and its comparison group



Mean minutes per e-mail run, days 1–9 against days 10–14

- Pilot S1–S3: 35.85 → 13.73 (**−22.12**) · comparison S4–S5: 34.67 → 34.10 (**−0.57**)
- A **comparison group** measures the effect of the period itself

4.4

Correlation is not causation

- r measures **association**, not cause
- A **confounding variable** influences both x and y
- Mondays: more mail **and** more interruptions
- Manual Monday runs: **37.6 minutes** · other days: **34.5 minutes**
- Interruptions and minutes may both depend on the day of the week

4.4

What 14 days and five people cannot show

- **Sample size**: at most 70 runs per routine; jarvis = 15 runs, 3 people, 5 days
- **Scope**: one studio, three September weeks — not “consultancies in general”
- **Measurement**: self-logged minutes, rounded; jarvis timed by the assistant
- **Novelty**: new tools are used carefully at first; effects can shrink
- The result **suggests**; it does not **prove**

4.4

Writing the interpretation: number, then limit

- Sentence 1 — the number, with **n** and the **period**
- Sentence 2 — the main **limit**

Example

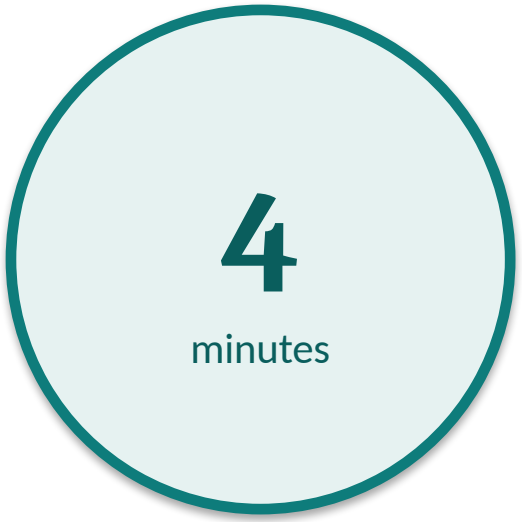
From day 8, daily-report runs took 10.0 minutes less on average (26.8 → 16.8; 35 runs in each group; **SD** (Standard Deviation) 4.4 and 3.1).
This before/after comparison covers two weeks in one studio, so it cannot separate the template from other changes made at the same time.

4.4

Activity 4 — fun quiz

⏱️ **0:56–1:00 · Fun quiz · 4 minutes**

- 1 — r = −0.9. What does it mean?
- 2 — r = 0.1. What does it mean?
- 3 — Does n = 14 “prove” a relationship?
- 4 — Early coffee drinkers triage e-mail faster. Does the coffee speed them up?



4.4

Real-life example 4.4 — did it work?

TASK

r for three pairs, the before/after table of the template, the pilot against its comparison group, and a two-sentence interpretation (number, then limit) for each.

Deliverables

Correlation table (pair, routine, n, r, wording)
Before/after and pilot/comparison tables
Four two-sentence interpretations in docs/stats_notes.md

Where it goes next

Brick 4 (Oct 22) — r, before/after, interpretations
Brick 7 (Nov 12) — correlation and linear_regression in StatsEngine
Brick 11 (Dec 17) — charts and the recommendation
Mid-Term Part A (Nov 20)

Demo in the lecture module: Start / Pause, Step, Restart — the trace window shows every executed line.

4.4

Videos for home — Lesson 4.4

12:18

Correlation Doesn't Equal Causation: Crash Course Statistics #8

CrashCourse · YouTube · Scatter plots, the direction and strength of r, and a third variable that causes both A and B.

7:20

Example: Correlation coefficient intuition | Mathematics I | High School Math | Khan Academy

Khan Academy · YouTube · Matching scatter plots to values of r by eye: direction and strength without a calculator.

10:45

Correlation and causality | Statistical studies | Probability and Statistics | Khan Academy

Khan Academy · YouTube · A headline that confuses correlation with causation, taken apart with possible confounding variables.

All three were confirmed working (oEmbed) on October 11, 2026, and are 15 minutes or shorter. Check a link with VirusTotal before you open it.

B4

Workshop — Brick 4 “Statistical Analysis Script”

time	what happens
1:00–1:10	Brief; open analysis/stats_sheet.py (csv + statistics only)
1:10–1:45	Descriptives per routine + one routine by hand
1:45–2:15	Text histogram; outliers with your stated rule
2:15–2:40	r for two pairs; before/after table; interpretations
2:40–3:00	Checkpoint: one surprising number per team; commit

B4

Brick 4 — what is submitted

- `analysis/stats_sheet.py` + output + `docs/stats_notes.md`
- One routine computed **by hand**
- `jarvis/stats_engine.py`: **signatures and docstrings only** (sample or population stated)
- **Formula Card v1** + one paragraph on your own 14-day log
- Weekly Progress Report #4

Due

Thursday, October 22, 2026
23:59 · **LMS** (Learning Management System) + GitHub

Phase 1 closes:
Bricks 1–4 complete

B4

Home work (2 hours)

- Finish **Brick 4**
- `stats_engine.py`: signatures + docstrings (formula; “sample (n − 1)”)
- **Formula Card** (one page, handwritten allowed): every formula of Classes 3–4
- Run `stats_sheet.py` on **your own 14-day log** and compare with the studio
- Leedy & Ormrod, Chapter 11, *Analyzing Quantitative Data* (20 minutes)
- Weekly Progress Report #4

B4

Next Friday — Class 5 (October 23)

- **Python & OOP (Object-Oriented Programming) Refresher + Design: Blueprint Before Bricks**
- Dataclasses for a time entry; a loader from **CSV** (Comma-Separated Values) to objects
- The **UML** (Unified Modeling Language) class diagram of JARVIS; abstract base classes for tools and providers
- Brick 5 — Design Pack + Working Loader — *summary only, created next week*

B4

Key references

- **Python documentation — statistics module** — PSF
- **e-Handbook 1.3.5.1 — Measures of Location** — NIST
- **e-Handbook 1.3.5.6 — Measures of Scale** — NIST
- **e-Handbook 1.3.3.7 — Box Plot** — NIST
- **e-Handbook 1.3.3.26 — Scatter Plot** — NIST
- **Statistics Canada — Statistics: Power from Data!** — Statistics Canada
- **Power from Data! 4.5.3 — Calculating the variance and standard deviation** — Statistics Canada
- **Microsoft Support — QUARTILE.INC function** — Microsoft
- **Pearson — Leedy & Ormrod, Practical Research: Planning and Design (12th ed.)** — Pearson (publisher)
- **OpenIntro Statistics (free textbook)** — OpenIntro
- **VirusTotal — URL scan** — VirusTotal (Google)
- **Cyber.SoHo Legal Notice, Terms of Use and Acceptable Use Policy (v2.0)** — Cyber.SoHo

The full list, each link with its explanation and a VirusTotal button, is in the lecture module and the hand-out.

B4

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Boréal Studio is a fictional teaching case; all data are synthetic. Quizzes are formative and never graded.

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